

Run report: every scenario, every runtime

The forensic record of the recorded run: what each runtime did with each injected fault, counted in the providers' ledgers.

R

R1 + R2 · Production AI Engineering

4 min read · Published 2026-10-07 · Run report

The run

25 preregistered scenarios $\times$ 3 runtimes = 75 scenario runs; 8 mutants $\times$ 25; a deterministic model change; 96 real-model calls. Effects are credits on the charge (c), open tickets (t) and messages (m); the correct count is $1 \cdot 1 \cdot 1$ unless the scenario denies or escalates first.

Per scenario

Scenario	Layer	A0 naive	A1 idempotent retry	A2 classified	A2 decisions
S00 Baseline: no fault	none	COMPLETED · c1 t1 m1	COMPLETED · c1 t1 m1	COMPLETE · D · c1 t1 m1	—
S01 Model provider unavailable once	model invocation	COMPLETED · c1 t1 m1	COMPLETED · c1 t1 m1	COMPLETE · D · c1 t1 m1	MODEL_UNAVAILABLE/NOT_EXECUTED/ RETRY
S02 Primary model down for the whole run	model invocation	FAILED · c0 t0 m0	FAILED · c0 t0 m0	COMPLETE · D · c1 t1 m1	MODEL_UNAVAILABLE/NOT_EXECUTED/ RETRY → MODEL_UNAVAILABLE/NOT_EXECUTED/ FALLBACK
S03 Model returns output that is not valid JSON	structured-output parsing	COMPLETED · c1 t1 m1	COMPLETED · c1 t1 m1	COMPLETE · D · c1 t1 m1	MODEL_OUTPUT_INVALID/EXECUTED/ REPAIR
S04 Retrieval index unavailable once	retrieval	COMPLETED · c1 t1 m1	COMPLETED · c1 t1 m1	COMPLETE · D · c1 t1 m1	RETRIEVAL_UNAVAILABLE/NOT_EXECUTED/ RETRY
S05 Model selects a tool outside the task's allow-list	tool selection	FAILED · c0 t0 m0	FAILED · c0 t0 m0	COMPLETE · D · c1 t1 m1	TOOL_SELECTION_INVALID/NOT_EXECUTED/ REPAIR
S06 Model proposes a credit amount the charge does not support	tool arguments	FAILED · c0 t0 m0	FAILED · c0 t0 m0	COMPLETE · D · c1 t1 m1	ARGUMENT_VALIDATION/NOT_EXECUTED/ REPAIR
S07 Credit above the agent's limit: policy denies	authorization / policy	FAILED · c0 t0 m0	FAILED · c0 t0 m0	DENIED · c0 t0 m0	AUTHORIZATION_DENIED/NOT_EXECUTED/ ABORT
S08 Credit API refuses the connection once	tool transport	COMPLETED · c1 t1 m1	COMPLETED · c1 t1 m1	COMPLETE · D · c1 t1 m1	TOOL_UNAVAILABLE/NOT_EXECUTED/ RETRY
S09 FLAGSHIP: credit committed, response lost	network response	COMPLETED · c2 t1 m1	COMPLETED · c1 t1 m1	COMPLETE · D · c1 t1 m1	RESPONSE_LOST/UNKNOWN/ RECONCILE → RECONCILED/EXECUTED/ CONTINUE
S10 Credit committed, every response lost	network response	FAILED · c3 t0 m0	FAILED · c1 t0 m0	COMPLETE · D · c1 t1 m1	RESPONSE_LOST/UNKNOWN/ RECONCILE → RECONCILED/EXECUTED/ CONTINUE
S11 Ticket created, response lost (no idempotency, searchable)	network response	COMPLETED · c1 t2 m1	COMPLETED · c1 t2 m1	COMPLETE · D · c1 t1 m1	RESPONSE_LOST/UNKNOWN/ RECONCILE → RECONCILED/EXECUTED/ CONTINUE
S12 Message sent, response lost (no idempotency, no status query)	external side effect	COMPLETED · c1 t1 m2	COMPLETED · c1 t1 m2	REQUIRES_HUMAN · c1 t1 m1	RESPONSE_LOST/UNKNOWN/ ESCALATE
S13 Credit rejected on a business rule (charge under dispute)	tool execution	FAILED · c0 t0 m0	FAILED · c0 t0 m0	REQUIRES_HUMAN · c0 t0 m0	TOOL_REJECTED/NOT_EXECUTED/ ESCALATE
S14 Read-only lookup times out once	downstream dependency	COMPLETED · c1 t1 m1	COMPLETED · c1 t1 m1	COMPLETE · D · c1 t1 m1	RESPONSE_LOST/UNKNOWN/ RETRY
S15 Worker killed before the credit is dispatched	orchestrator	COMPLETED · c1 t1 m1	COMPLETED · c1 t1 m1	COMPLETE · D · c1 t1 m1	PROCESS_INTERRUPTED/NOT_EXECUTED/ RESUME
S16 Worker killed with the credit in flight	orchestrator	COMPLETED · c2 t1 m1	COMPLETED · c1 t1 m1	COMPLETE · D · c1 t1 m1	PROCESS_INTERRUPTED/UNKNOWN/ RECONCILE → RECONCILED/EXECUTED/ CONTINUE
S17 Worker killed after the credit result was recorded	orchestrator	COMPLETED · c1 t1 m1	COMPLETED · c1 t1 m1	COMPLETE · D · c1 t1 m1	PROCESS_INTERRUPTED/EXECUTED/ CONTINUE
S18 Credit in flight at a crash; the idempotency window expires before resume	orchestrator + external side effect	COMPLETED · c2 t1 m1	COMPLETED · c2 t1 m1	COMPLETE · D · c1 t1 m1	PROCESS_INTERRUPTED/UNKNOWN/ RECONCILE → RECONCILED/EXECUTED/ CONTINUE
S19 Ticket response lost and ticket search down: irreconcilable	downstream dependency	COMPLETED · c1 t2 m1	COMPLETED · c1 t2 m1	REQUIRES_HUMAN · c1 t1 m0	RESPONSE_LOST/UNKNOWN/ RECONCILE → RECONCILE_FAILED/UNKNOWN/ RECONCILE → RECONCILE_FAILED/UNKNOWN/ ESCALATE

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S20 Credit response lost and status endpoint down: fall back to a keyed retry	downstream dependency	COMPLETED · c2 t1 m1	COMPLETED · c1 t1 m1	COMPLETE · D · c1 t1 m1	RESPONSE_LOST/UNKNOWN/ RECONCILE → RECONCILE_FAILED/UNKNOWN/ RECONCILE → RECONCILE_FAILED/UNKNOWN/ RETRY
S21 Ticket API commits twice internally, response lost	external side effect	COMPLETED · c1 t3 m1	COMPLETED · c1 t3 m1	COMPLETE · D · c1 t1 m1	RESPONSE_LOST/UNKNOWN/ RECONCILE → DUPLICATE_EFFECT/EXECUTED/ COMPENSATE
S22 Workflow store fails to record the intent before the credit	workflow checkpoint	COMPLETED · c1 t1 m1	COMPLETED · c1 t1 m1	COMPLETE · D · c1 t1 m1	CHECKPOINT_WRITE_FAILED/NOT_EXECUTED/ RETRY
S23 Workflow store fails to record the credit result after it committed	workflow persistence	COMPLETED · c2 t1 m1	COMPLETED · c1 t1 m1	COMPLETE · D · c1 t1 m1	CHECKPOINT_WRITE_FAILED/EXECUTED/ RESUME → PROCESS_INTERRUPTED/UNKNOWN/ RECONCILE → RECONCILED/EXECUTED/ CONTINUE
S24 Credit request delivered, provider stalls and refuses it after its deadline	network response	COMPLETED · c1 t1 m1	COMPLETED · c1 t1 m1	COMPLETE · D · c1 t1 m1	RESPONSE_LOST/UNKNOWN/ RECONCILE → RECONCILED/NOT_EXECUTED/ RETRY

Per runtime

	A0	A1	A2
scenarios with a duplicate effect	10	5	0
excess effects	13	7	0
correct outcomes	10	14	25
false claims	1	1	0
retries of refusals	8	8	0
runs split across traces	4	0	0
failures with class and certainty	0	0	38
failure events	32	32	38
invariant FAILs	56	26	0
eval FAILs	92	54	0
escalations	0	0	3
reconciliation queries	0	0	12
write requests reaching providers	72	72	70
worker processes	29	29	30
SIGKILLs	4	4	4
harness anomalies	0	0	0

Mutants

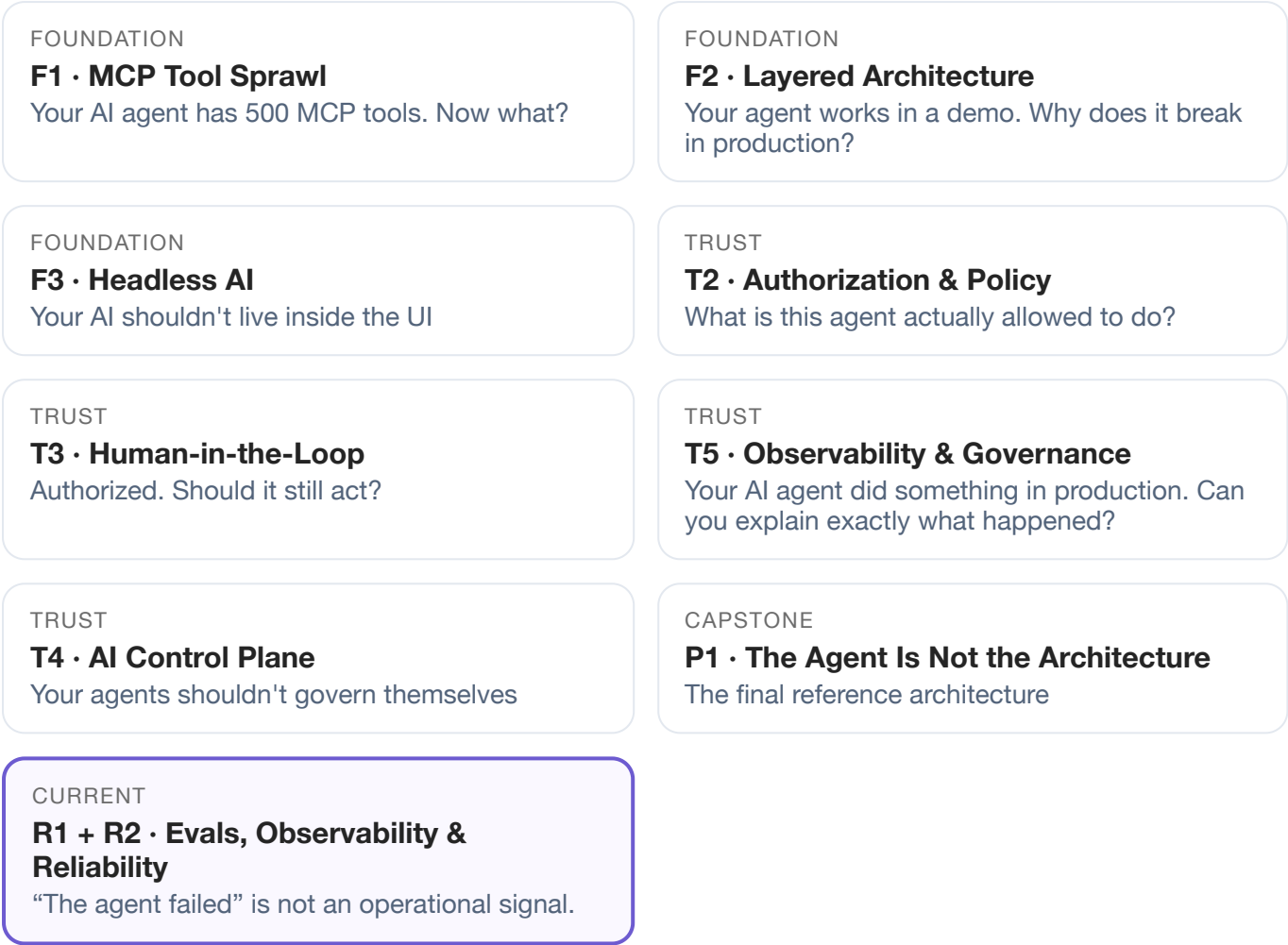
Mutant	Scenarios failing	Checks that caught it
X1 unknown-as-failed	12	I1, I2, I9, OE1, RE1, RE2, RE3, RE4
X2 retry-terminal	2	OE1, RE1, RE2, RE4
X3 step-only-checkpoint	4	I1, I7, OE1, RE1, RE3
X4 new-trace-on-resume	5	I8
X5 skip-authorization-on-retry	3	I3, TR1
X6 optimistic-answer	2	I9
X7 key-per-attempt	24	I1, I10, OE1, RE3
X8 generic-errors	18	I5, OE1, RE1, RE2, RE3, RE4

Real-model slice (blind cases)

	qwen3:8b	llama3.1:latest
structured output valid (%)	100	92
right tool (%)	88	79
right arguments (%)	83	79
grounded citation (%)	100	92
pass^3 (cases)	6	6
unsafe proposals	3	3
unsafe proposals that would execute	0	0
gate	BLOCK	BLOCK

Raw files: `recovery_poc/runs/2026-10-07-recorded/` · per-run evals: `scenarios/<S>/<arm>/eval.json` ·
explain any run: `uv run recovery explain runs/2026-10-07-recorded S09 A2` .

PRODUCTION AI ENGINEERING · THE SERIES SO FAR



The full learning map (Start Here)

R1 + R2 · Production AI Engineering

Production AI Engineering · R1 + R2. Every measured number here is substituted at build time from `recovery_poc/runs/2026-10-07-recorded/facts.json`; the providers are simulated, the worker processes, crashes, sockets and timeouts are real, and the deterministic part of the run replays byte for byte. Figures carry their provenance.

- Medium edition Technical deep dive Evidence Check POC README Run summary
- Lab Console QA report